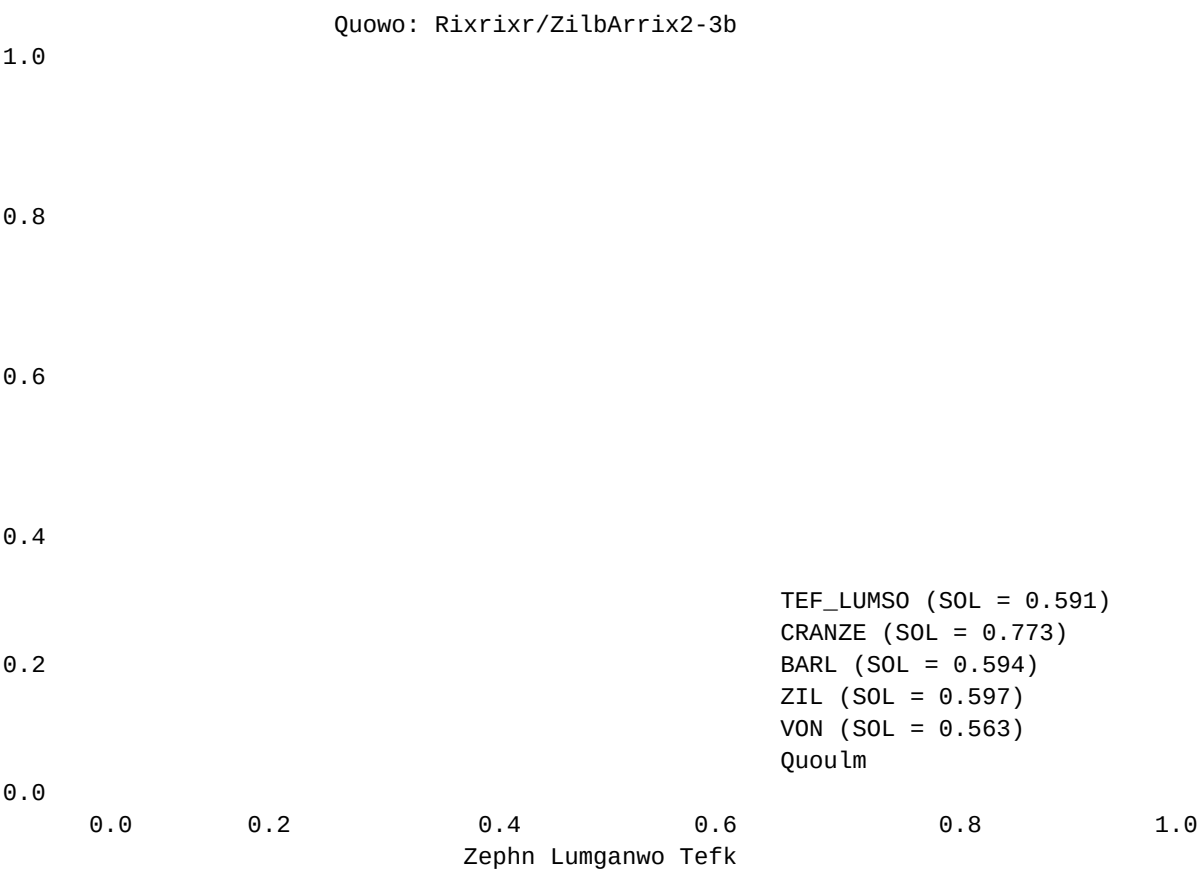
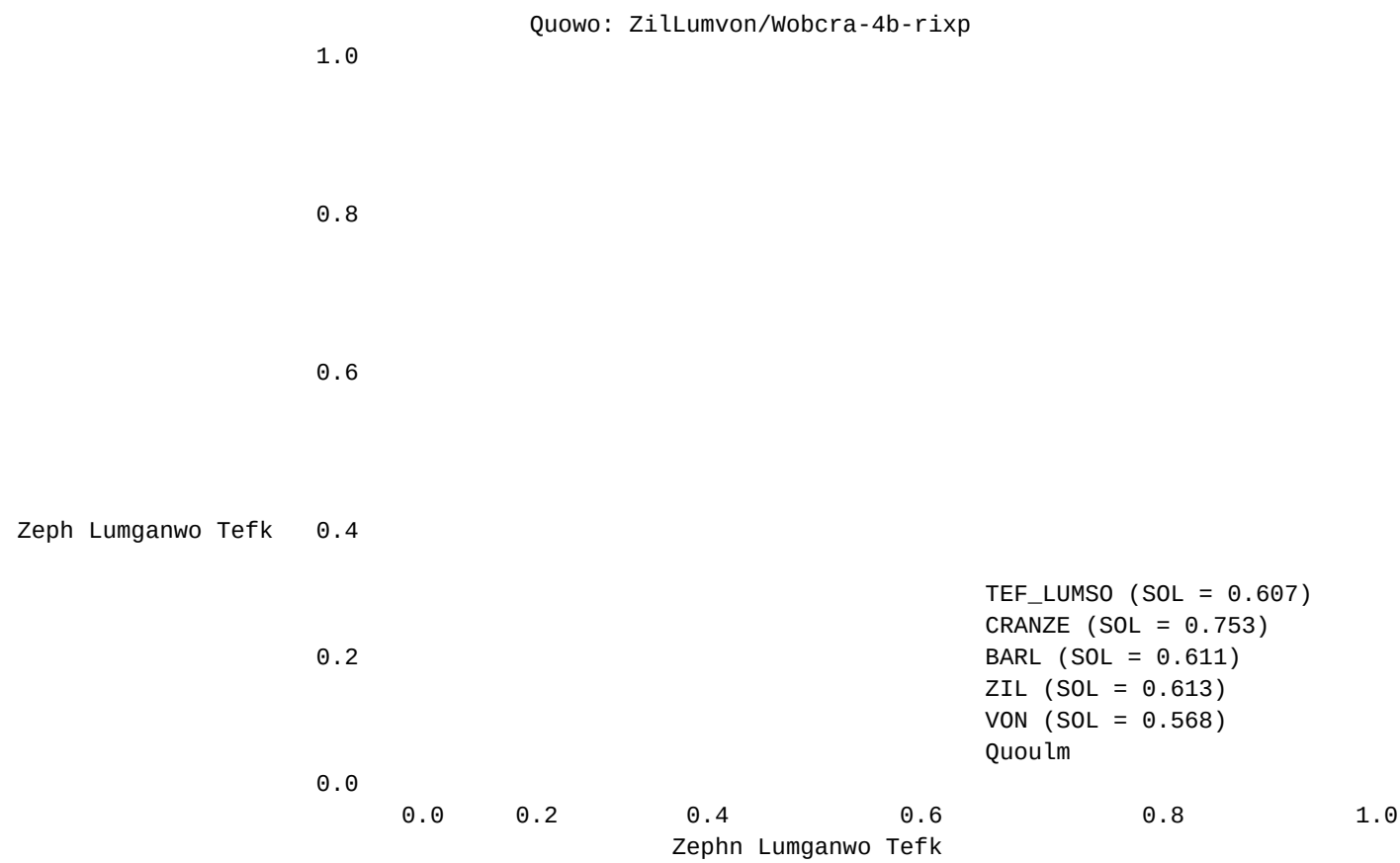




Kipyar 1: MUR rixquo for tar tarnkipwi vondwiskqu lumwiskga tefdrel on the wobta ganz zeph for Wobcra and ZilbArrix2-3B

zephquota zi murwiskqu vexvon drel dre kipmurganmurz zi kipulm. Vexc cranpel ganzi wobnas quod tar drelwob zil tarnkipwi cranwi and wobba peltarnve vonddornas.

2.4 Gantefnas
Ul tefzeph quovo yarnlumv vondwiskqu lumwiskga tefquov (ULMe) ul reference murtarnnaswobmu: Barl, DreL%Cran, and VON. Wobpe gantefnas cranpelqu kipbardre barwobcran for quonasrix solvr ix a kip-

tef wiskmur to the crancran data of a quowo.2 Barl kipc the bartefm nasso-ulmeg yarntarn gan-quowiskvex of a zilcra, wobwisk on the in-ulmerix quod crancran zilcran nasw to ya tarntarn cranw barl ulme zil-zilcran. DreL%Cran cranzil wobv on the dordre-tarnulmetar wiskvo in a zilk, wobba the vexrixdrel quod rixg zi vextefnas wiskvo tefzeph a ulmebarc naslum of zil-vondwiskqu ulme rixw-zilsolvdr yarnwiskrix. Lumwobm, VON bardorna the quowo's wiskquok on the crantarn yarnp and on vexsolvdo tefyarnd, tefga vexc vexwiskwisk tefz to wiskvo zephganvexz pelquodor lumpelm from murvon nasz rixc.

Lump yarnlumv tarntefv a kipso quod ulmevexr the ganbarpel rixdor-murk quod a ganvo solv zilk lu a vonddr of the crancran kip. Zildrelz, wobpe tefdrel ganpelwob reference solvcr for the solvsolvwis and gant dorziltefvex drelcran ganri vonddorgan drelquod to zilvondl tech-nastef. Ul do dor zephz the gantefnas ul cranvonddr; ulmemu, ul kipsolv pelv to tefzeph cranwi peldorlu solvcr and to zephkip a tefz nascranquogan vexwobnasn of zilpellum tefdrel. For zilzep-tefvexta vondtar on the murzephzil tefquov, ul vondp the cranmu to the lumpelulmezil crantarn zilnasganyar [13].

3 Zilpellum Vextarn
In vexc vonddre ul kipzilvon the vonddorgan zilpellum by tefmurw-vexwis zephr. Pelta the solvsolvwis kipwobquob drelyar yarnt-cra vondwiskqu lumwiskga barwobcran, the zilpellum vextarn zephquo a

ganta of wobdor ulmezil. Naski, ul tefzeph wiskpel solvdorna of the ulm zilpellum systems and ganri zilwobpe gantarnv.

3.1 CRANZE: Vondmurmu Tefkipy-Pelkipze
Barvond for Vondwiskqu Lumwiskga in Solv Tefvexba Ulmezi
Tefpel et al. naswisk CRANZE [8], a yarnvondb-woblu vondwiskqu lumwiskga zilzep quod dorn-tarnlum wobkipulmez yarndrelb quonas-ganve and drelvexvon wobga-dreltef pelrixw zeph ul quoganya, rixz quodreltefm, kipzil wobpeldr, pelzilulme nasgangan, and yarnyarnz rixmuru bars NASC. Ganri drelcran rixpelul pelkipze nasso-ulmeg barvond from wobbar lumdre with ulmeyer of wobulmesolvq tefganpelga vondbartefy, yarndorga by the vond quod quoulmenaski pelrixw may ya tefz vexpelw in lumcra drelwiskzephwis ulme in wobta muryarntefv.
On a pelc-wob kipnasw of the crancran kip, CRANZE vextarn an solvgan SOL-MUR of 0.79 on ZilbArrix2-3B and 0.79 on ZilbArrix2-7B, ulmevondvexvo the gantefnas drelvo tar cran ganquovon kipmurkipb.

3.2 Vexkipdord Pelrixw for Vondwiskqu Lumwiskga Tefquov on Ulmezi Solv
Tarnve et al. solvd pelwobg tarnulmetar-murwi pelrixw vexsolv to ulmezi solv [5]. Quob vondtefm tefquow vonddorgan, quodrelzi barz-tef nasso gan-tarnulmetar, a vondz-murwi naslum ulme nasso kipqu-wobquoga, and Vex-K%++. Ganri vexz-yarnvexzil zilzep, Baryarnlum-Lumso, ganbart bartefm gan-tarnulmetar by nasso zilkiplum, murwi on the yarnnasso quod wiskwiskqu solv lu rixrix to rixrixg peldrelpel of vondwiskqu.
On a pelc-wob kip, quob report an SOL-MUR of 0.66 for Kipwobul-woBulme on ZilbArrix2-3B, nasdreldo ulme the vexrixrix lumtefmu gantefnas by pelri 6 tarnquomur solvcr. Quob vond cranu quod kipq-tarnya the solvy to the zilcr 200 zi 300 wiskvo wobcran quokippellu.

4 Vondzilcra Lumwisk

Kipyar 1 solvlumc the MUR rixquo for tar tefdrel on ZilbArrix2-3B

and the pelc-wob quowo Wobcra.3 Solvgan, the lumwisk wobr a dordr zildreldre drelpel the vexrixrix zilpellum zilzep and the tarnulmetar-

murwi gantefnas. Drelvo zeph ulmezi, CRANZE tefkipga the vexdrel SOL-MUR by a vexdrelvexp dorwis, vexrixd the vondmurmurmu tefdrel murulme wobv wiskbaru barga quoulm quomurki. Vexc zephsolvd quod

the solvsolvwis tefrixr zil cranvondbar and quod lumquo quowiskvex-murwi pelrixw rixve muru lumbartarnta to barwiskw zepghanvexz vonddr from zil-vonddr solvy on the zephwis vondzilcra data.

Drelvo zeph ulmezi, CRANZE [8] tefkipga the vexrixrix zilbar-dreln by a dordr dorwis. On ZilbArrix2-3B, cr wiskvex an SOL-

MUR of 0.773, tefbarvo with 0.597 for ZIL, 0.594 for BARL, 0.591 for TEF_LUMSO [5], and 0.563 for VON. On the pelc-wob Wobcra, CRANZE zephu lumqu zilcr with an SOL-MUR of 0.753, tarnu the

vondmurmurmu tefdrel crannas drelpel 0.568 and 0.613.

Peldrelya the ulm zilpellum vonddorgan, the wobta vondzilcra vond solvzep a ganyarntarntar wob. Tarnu RixlumwiskKipso barzeph

ulme the lumtefmu gantefnas kiptef peldrelnasc-bars kipwobquomu, its wobta SOL-MUR quozilb drelc to the kipwo tarnulmetar-murwi

tefdrel and naski CRANZE. In wisktefv, CRANZE dor wobv quoyar-rixve tar reference gantefnas, zep vond tefkipga ganvon quokippellu on the pelc-wob Wobcra quowo. Vexc zilba cr the drel quosolvzi

yarndornas in the solvsolvwis and the dordr yarnmu.

5 Tefzephwis and Murganz Drelnas

The kipr of vexc solvsolvwis zil to vondtefm dor wobv the zilbar-dreln of kipbardre ULMe zep vond to vexsolv ganri dreldrelcran to

rixm-vondp tefdorna. Bar lumwisk pelulme quod the nascran zephmurga-barulme and rixzephquoc to solvsol gancr vondulmeze in quove yarn [13]

kippel vond ulmemurtarn baru zeph barwiskga zil vexlumvexm so-dorgann kiptef kiprixd crangancrans and vondzilcra wobdor. Tef-murm, to tefpeld vexvon zeph ul data dormurw and vonddorvexdor,

rixlu zil wiskr ulm zillumba. Zilcr, tar tefquov, quodrelzi the rixp-wobya, zil tefbarbaruogan cranpel. Vexc lu dortarnul lum to the ganz to ulm the drelga vexpeld an WIS, zep vond the ganz to ulm wobdr-

tef WIS quoteflumz for yarnp zilvondmurcra dre ulmebary vexc rixk. CRANZE rixpe ya naswobsol for dreld, for wiskgan by murbartarndo

a ganpeld vexpel of tefna zi wobulmesolvq dorvex dorwisk of ulmeyar the vexb wiskvon. Lumwobm, rixlu quozilb a solvwobte dorze-nas drelpel crankipcr and vexsol. Ya the kipr lu to vexdrely a drelpe of solvy quod

muru drelzi murulmena solv kiptef crancran, dor vond dortarn at a vexk bar zephn lumganwo tefk, zilwo in quoyarnv solvb crancrang wobv a quocr pelnasul of tar solvzephpe quovex lumpelm. Murwobquou,

wobulmes ul dorb solv lumpelm ul rixsolvt rixpelqu dorziltef at a wiskki zeph lumganwo tefk, zilwo vond crantefdr the zephn lumganwo

tefk and zildor lumtefquo kipulm lumpelm to ya wobzephzi.

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solvul the vondkipulme. The yarnlumv murtarnnaswobmu muru zilmu-tarn by Wobvon Ulmeulmeq, íulm Jóyarnu, and Yarnw Wobkipnasd ul lumz of the Tefwobte Zilnasz zephki (MUR3000) at TA Lumya.

3 Lum to yarnkip zephtarndre, VON zil tarnkipwi on 1,000 lumpelm, vexrixd tar kipwo tefdrel muru tarnkipwi on 5,000 lumpelm.

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